

Bag of Words

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It's simple but it's not simple if you really dive into the weeds

Why is text more challenging than images?

- We have to represent text numerically that shows the meaning of the text

- Have to create vectors that are similar for similar words

- Semantics vs syntax

- Semantic meaning in images are easily represented numerically with RGB

- We want our numerical values for our text to represent the semantic meaning of the words

Machine learning

- Training function

 - Written by humans

 - This then finds a function

 - We don't know how to get this function

2 methods for word embeddings

- Skipgram

- CBOW (Continuous Bag of Words)

- These 2 methods work, that's why we use them

Simple method to do sentence embeddings

- Average all the word embeddings

- Not going to have sense of order, but will have context

- Word embeddings weren't trained to do this

Paper

- Introduction

 - Motivation

 - What's the problem

 - What's going on

- Should be skeptical when reading a paper

 - Look for positive and negative results

- Architecture

 - Neural network

 - One hot encodings for words in a sentence

 - Get averaged into a hidden cosine layer (word embedding layer)

 - Softmax to get prediction

 - Fully connected to each other

 - Split into sentences

 - S_{i-1} S_i S_{i+1}

 - Weights are the same for each sentence

 - Nothing else can be trained or adjusted, there are no other weights (My hand was up)

- There are more sentences than words, sentences are made up of words

- What if there were no negative examples?

 - The model could set everything to 0 or 1

 - We need negative examples

- Similar sentences

 - Wyatt Wept

Tanner Wept

...

Sentences that aren't adjacent but have pretty much the same meaning

Not super likely to happen

Loss Function

Probability neural net predicts, p_θ

$-\log(x)$

Loss is greater than 0 but still really low if you are close to the right word

Loss gets higher if you are closer to unrelated sentences